

AI Business Analytics Dashboard Project Report

Project Title:

AI Business Analytics Dashboard Using Python, Stream lit, SQLite, and Machine Learning

1. Introduction:

The **AI Business Analytics Dashboard** is a web-based business intelligence application developed using **Python** and **Streamlit**. The dashboard enables organizations to analyze sales data, monitor key business performance indicators (KPIs), generate reports, and predict future sales revenue using a machine learning model.

The system also provides a secure authentication module with **Login, Registration, and Forgot Password** functionality using SQLite and encrypted passwords.

2. Project Objectives:

The main objectives of this project are:

- Develop a modern business analytics dashboard.
 - Provide interactive sales data visualization.
 - Display important business KPIs.
 - Implement secure user authentication.
 - Predict sales revenue using Artificial Intelligence.
 - Generate downloadable business reports.
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3. Technologies Used:

Technology	Purpose
Python	Programming Language
Streamlit	Web Application Framework
SQLite	Database Management
Pandas	Data Processing

Technology	Purpose
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Plotly Express	Data Visualization
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Scikit-learn	Machine Learning
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Joblib	Loading ML Model
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OpenPyXL	Excel Report Generation
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FPDF	PDF Report Generation
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Bcrypt	Password Encryption
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4. Project Structure:

AI-Business-Analytics/

| — app.py

| — login.py

| — register.py

| — forgot_password.py

| — prediction.py

| — report.py

| — database.py

| — analytics.db

| — requirements.txt

| — assets/

| | — logo.png

| | — style.css

| — models/

| | — sales_model.pkl

└─ data/

└─ sales.csv

5. System Features:

User Authentication:

The application provides a secure authentication system with the following features:

- User Login
 - User Registration
 - Forgot Password
 - Password Encryption using Bcrypt
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Dashboard Features:

The dashboard displays important business KPIs including:

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value

Users can filter data by Region and Category for detailed analysis.

Data Visualization:

Interactive charts available in the dashboard include:

- Revenue Trend
- Revenue by Category
- Revenue by Region
- Top Selling Products

These charts help users quickly understand business performance.

AI Sales Prediction:

The dashboard includes an Artificial Intelligence module that predicts future revenue based on user input.

Input Variables:

- Quantity
- Price

Output:

- Predicted Revenue

The prediction model is stored as:

models/sales_model.pkl

6. Database Design:

Database Name:

analytics.db

Users Table:

Field	Data Type
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id	INTEGER
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username	TEXT
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password	TEXT
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Sales Table:

Field	Data Type
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OrderID	INTEGER
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Date	DATE
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Customer	TEXT
----------	------

Product	TEXT
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Field	Data Type
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Category	TEXT
----------	------

Quantity	INTEGER
----------	---------

Price	REAL
-------	------

Region	TEXT
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Total	REAL
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7. System Workflow:

1. The user opens the application.
 2. The user logs in or creates a new account.
 3. Sales data is loaded from the SQLite database.
 4. Dashboard KPIs are calculated.
 5. Interactive charts are displayed.
 6. Users can filter sales records.
 7. AI predicts future sales revenue.
 8. Reports can be downloaded in CSV, Excel, or PDF format.
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8. Report Generation:

The system supports downloading reports in multiple formats:

- CSV Report
- Excel Report
- PDF Report

These reports allow users to share and analyze business data outside the application.

9. Advantages:

- User-friendly interface
 - Secure authentication system
 - Interactive business dashboard
 - Fast data visualization
 - AI-powered revenue prediction
 - Multiple report formats
 - Professional and responsive design
 - Easy database management
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10. Future Improvements:

Future versions of the project may include:

- Email verification
 - Multi-user roles (Admin/User)
 - Cloud database integration
 - Real-time sales monitoring
 - Inventory management
 - Customer analytics
 - Advanced AI forecasting
 - Mobile responsive interface
 - Dark and Light theme support
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11. Conclusion:

The **AI Business Analytics Dashboard** is a complete business intelligence solution developed using Python, Streamlit, SQLite, and Machine Learning. It provides secure user authentication, interactive data visualization, AI-based sales prediction, and report generation. The application enables businesses to monitor sales performance, analyze KPIs, and make informed decisions through an intuitive and professional dashboard.

Project Name: AI Business Analytics Dashboard

Developer: Ahmed Masood

Technology Stack: Python, Streamlit, SQLite, Pandas, Plotly, Scikit-learn, Joblib, OpenPyXL, FPDF, Bcrypt
